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$$\begin{aligned} & \lim_{x \rightarrow 0} \left(\frac{ax + b}{cx + b} \right)^{\frac{1}{x}} \\ & \text{Stel } t = \frac{1}{x} \\ & = \lim_{t \rightarrow \infty} \left(\frac{at^{-1} + b}{ct^{-1} + b} \right)^t \\ & = \lim_{t \rightarrow \infty} \left(\frac{bt + a}{bt + c} \right)^t \\ & \text{Stel } s = bt + c \\ & = \lim_{s \rightarrow \infty} \left(\frac{s - c + a}{s} \right)^{\frac{s-c}{b}} \\ & = \lim_{s \rightarrow \infty} \left(1 + \frac{a-c}{s} \right)^{\frac{s}{b}} \cdot \lim_{z \rightarrow \infty} \left(1 + \frac{a-c}{s} \right)^{\frac{-c}{b}} \\ & = \left(\lim_{s \rightarrow \infty} \left(\frac{s - c + a}{s} \right)^{\frac{s}{a-c}} \right)^{\frac{a-c}{b}} \\ & = e^{\frac{a-c}{b}} \end{aligned}$$

References